

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

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6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a food-security floor of 90% of open-field yield with a differentiable simulator of solar geometry, trackers, PV output, crop light, microclimate, soil water and the SIMPLE crop model. The crop response to shade is the least certain part. A hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, and we turned the learned curves and four field trials into an ensemble of plausible responses by gradient descent through the simulator. In a deployable protocol (density chosen from each site's history, held-out sites and years, 400 crop-site pairs), light-sharing rules allowed arrays three times as dense as static arrays at equal retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24; a phenology rule needing no forecast was the best. A neural controller trained through the simulator exceeded the written rule by 0.044 LER but a rule tuned on the same training seasons by only 0.018 (95% interval 0.012–0.024); training on the ensemble or adding a farm-measurable biomass signal changed nothing. The design decision mattered more: a density chosen under the mean response left about 70% of crop-site pairs at the floor when another plausible response held (oracle 85–87%), and a floor raised by 0.02 under the mean response did better at equal compliance than a rule built on the ensemble's quantiles. The simulator is too mild for maize and soybean, so gains for these crops are probably overestimated. Equipping 5% of staple cropland would give about 14 PWh yr⁻¹ but needs about 10 TWp of modules.

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8 Crop modelling, Differentiable simulation, Hierarchical shrinkage model

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly
36 require AV systems to retain most of the reference crop yield. The German
37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
39 floors. How much electricity can be produced without sacrificing food secu-
40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from

generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which is sample-inefficient and hard to scale to global, multi-decadal assessments.

Two questions follow. First, how much of the electricity that an agrivoltaic system can produce under a food-security floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-

analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but too mild for maize and soybean, so the gains for those two crops are probably overestimated (Section 3.7).
4. A learned controller adds a small gain over the best rule: 0.044 LER over the written phenology rule and 0.018 over a rule tuned on the same training seasons. Training it on an ensemble of shade responses, or giving it a farm-measurable biomass signal, does not change this (Section 3.8).
5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen under the mean response leaves about 70% of crop–site pairs at the floor when another plausible response holds (oracle 85–87%). A rule built on the quantiles of an ensemble of responses raises this by 9 points for 0.04 LER, but a floor raised by 0.02 under the mean response does better at equal compliance, so the ensemble is useful to size a margin and not to replace it (Section 3.8).
6. At the yield floor, agrivoltaics on 5% of the world’s staple cropland

could supply about 14 PWh yr^{-1} , but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.10).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean and potato. Harvested areas (rainfed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop-site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below -8°C , in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels $+1.5$, $+2$ and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.5.

2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

153 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 154 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 155 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 156 on the horizontal is the difference between the two. Solar position follows
 157 the standard declination and hour-angle equations.

158 *Array geometry and PV electricity.* We consider elevated horizontal single-
 159 axis trackers with north–south axes. The axis height is 1.25 module widths,
 160 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 161 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 162 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 163 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 164 rows when the modules are horizontal, which is the stow position during field
 165 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 166 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 167 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 168 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 169 to the sun, which minimises the shadow on the crop. Given a light-sharing
 170 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

171 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 172 ule, reduced by the fraction of the module shaded by the neighbouring row,
 173 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 174 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
 175 with an hourly air temperature interpolated sinusoidally between the daily
 176 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 177 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 178 losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A con-
 179 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 180 energy reference E_{PV} .

181 *Validation of the PV model.* We compared the simulated specific yield of an
 182 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 183 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 184 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 185 returned data). Across sites, the simulated and PVGIS specific yields were
 186 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean

absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and with the different irradiance databases (NASA POWER vs. SARAH2, ERA5 and NSRDB). Electricity estimates are therefore, if anything, conservative.

Light and microclimate at the crop. Averaged over the row pitch, the fraction of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop’s shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$ for all methods (Section 3.2). Nocturnal warming under the modules is not represented. The reference evapotranspiration under the modules, ET_0 , is recomputed with the FAO-56 Penman–Monteith equation (Allen et al., 1998) from the reduced radiation and canopy temperature. Soil water conservation under the modules (Hassanpour Adeh et al., 2018; Elamri et al., 2018) therefore emerges from the model, but rain redistribution by the modules (Marrou et al., 2013a) is ignored.

Crop growth. Crop growth follows the SIMPLE model (Zhao et al., 2019) with the species parameters of its Table 1a (Table 1). Daily biomass growth is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

where $R_{\text{crop}} = s R$ is the radiation received by the crop. The terms are as follows.

- f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends only on thermal time, with senescence accelerated by heat and drought stress. Shade therefore reduces the daily biomass increment but not leaf-area development.

- f_{Heat} decreases linearly between the heat threshold T_{heat} and the extreme temperature T_{ext} of the canopy maximum temperature.
- $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for drought (Woli et al., 2012) is computed from a daily soil-water balance with curve-number runoff, drainage and root water uptake. Irrigated fields have $f_{\text{Water}} = 1$.

SIMPLE has a fixed harvest index and therefore ignores the high sensitivity of grain number to shading around flowering (Fischer, 1985; Andrade et al., 1999). It also ignores the differences in shade tolerance between crop types. In the meta-analysis of Laub et al. (2022), maize and grain legumes lost yield strongly even at low shade levels, whereas C3 cereals and tuber crops initially lost less than proportionally; shading trials also show marked effects of shade on maize development (Ramos-Fuentes et al., 2023). We added a stage-dependent penalty. During a critical window ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$ is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al. (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice 0.5, for which no specific curve was available) and lowest for potato (0.2). These values are assumptions that reproduce this ranking; they are not calibrated estimates. Section 3.6 shows the resulting shade-response curves. Because RUE is linear, SIMPLE does not credit the higher light-use efficiency of shaded canopies under more diffuse light. The penalty, conversely, makes shading around flowering more damaging than proportional. The net effect is closer to the field evidence than either assumption alone.

The thermal-time requirement to maturity was calibrated for every site as the mean thermal time between the GGCM sowing and maturity dates under the 2001–2020 climate, and the canopy parameters were scaled accordingly. The cultivar is thus adapted to the present climate, and warming shortens the season. Growth stops smoothly once the thermal-time requirement is met. Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site is represented by a rainfed and an irrigated sub-field in proportion to its SPAM irrigated fraction. Each sub-field carries its own array, so both yield and electricity are area-weighted over the two sub-fields.

254 2.3. SALS: simulation-trained adaptive light sharing

255 SALS learns a *control network* $u_t = \pi_\omega(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 256 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 257 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 258 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 259 are the information available on day t :

- 260 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 261 terception f_{Solar} , relative soil water and the previous day's ARID;
- 262 • a *forecast* of the day's maximum temperature (relative to T_{heat} and
 263 T_{opt}), irradiance and ET_0 ;
- 264 • static information: the irrigation status of the sub-field, the array's
 265 GCR g , latitude, day of season and the crop.

266 Forecasts are perturbed with random errors, independently every day: Gaus-
 267 sian with a standard deviation of 2°C for maximum temperature, and mul-
 268 tiplicative with standard deviations of 20% for irradiance and 15% for ET_0 .
 269 The physics always uses the true weather. The same forecast errors are
 270 applied to the rule-based controllers. The controller is conditioned on g .
 271 During training, each season is simulated with a GCR close to the density
 272 that is sustainable at its training site. This is the largest value on a coarse
 273 grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention
 274 of the site's training seasons at or above ρ , plus a uniform perturbation in
 275 $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to con-
 276 trol the densities that design selection will later pick. Outside the cropping
 277 season the trackers backtrack for maximum electricity.

278 For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the
 279 yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional
 280 PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

281 which combines an exact (hinge) and a quadratic penalty for the food-security
 282 floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the
 283 exact gradient of Eq. (3) is obtained by back-propagation through the 365
 284 daily steps of every season (back-propagation through time). This is the key

285 difference from reinforcement learning with black-box simulators (Gautron
286 et al., 2022), which estimates gradients from sampled returns.

287 Each minibatch contains 512 site-seasons, each simulated as a rainfed and
288 an irrigated sub-field, drawn from one of the four climates in turn. Seasons
289 with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training
290 and evaluation. The network is fine-tuned through the simulator with Adam
291 (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm
292 clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in
293 a numerically stable form, so that gradients remain finite even under extreme
294 heat.

295 *Initialisation.* Trained from random weights, the controller converged to
296 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
297 the seasonal-sharing rule, because the gradient of electricity with respect to u
298 is small and of the same sign for most days of a season. We therefore initialise
299 the network by behaviour cloning of the phenology rule of Section 2.4: binary
300 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
301 1.7 million day-states visited by the rule in the training seasons, with 1500
302 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
303 policy away from this initialisation wherever doing so improves the objective.
304 Consequently, the gain of SALS over the phenology rule measures what fine-
305 tuning through the simulator adds to that rule, not what can be learned
306 without it.

307 2.4. Deployable design selection and baselines

308 A real array has one fixed density, which must be chosen before it is built.
309 Every method, SALS included, therefore uses the same *history-based design*
310 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
311 for which the method’s mean yield retention over the site’s 2001–2014 sea-
312 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
313 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in
314 at least 80% of the historical seasons. The historical seasons of the test sites
315 are never used for training, and the weather of the evaluation years is never
316 used for design. We compare four controllers:

- 317 • *AV static*: backtracking trackers without light sharing ($u = 0$);
- 318 • *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the
- 319 cropping season;

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

- 320 • *Stress rule*: an expert rule that shares light ($u = 1$) during the main
321 growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress
322 forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules
323 track the sun and shade the crop;
- 324 • *SALS*: the learned controller.

325 The open field and a conventional PV plant without crops ($\text{LER} = 1$ each)
326 are the references. Because the design rule is identical for all controllers, dif-
327 ferences between methods reflect the quality of control and the array density
328 that each controller can sustain under the food-security floor. We also com-
329 pare the controllers at identical GCR over the whole grid (matched-density
330 frontiers).

331 2.5. A shade-response model learned from field data

332 The simulator needs a crop response to shade that is supported by ob-
333 servations. We learned one from the open data of the meta-analysis of Laub
334 et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observa-
335 tions of yield relative to an unshaded control at relative shade reductions of
336 0–90% from 58 studies, with crop and crop type, experiment type (artificial

shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

2.6. Shade-response uncertainty: gradient calibration, robust controllers and design

The crop’s response to shade is the least certain part of the simulator, so we treat it as an unknown with a distribution. Two parameters per crop enter the simulator: a radiation-use compensation c (growth is multiplied by $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop) and a multiplier h on the flowering-window harvest-index penalty. Because the simulator is differentiable, both can be fitted by gradient descent through whole simulated seasons to published yield–shading observations. The observations are the relative yields at 20% and 40% shading of the data-driven curves of Section 2.5 for the five crops (median, with an error equal to the width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice takes the curve of C3 cereals, which contains no rice), the field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) (each season of each trial as a separate observation, error 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. An earlier ensemble, E1, was fitted in the same way to curves read by hand from the figures of Laub et al. (2022) and was used to train the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why a response milder than the data cannot be lowered further. We fitted 12 ensemble members by randomised maximum a posteriori estimation:

every member sees the observations perturbed by their errors and a prior draw perturbed by its width, and is optimised with Adam (40 steps, with the loss still falling slightly at the last step; 40 training-block seasons per crop and member; shading of the static arrays cached). The ensemble is a device for generating plausible responses and not a validated posterior: it has no coverage check, and for some parameters (the harvest-index multiplier of wheat and potato) its spread equals that of the prior. The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The ensemble is used in three ways.

Robust and adaptive controllers. SALS-robust has the architecture and training of SALS (Section 2.3), but every training season is simulated with the response of a randomly drawn ensemble member that the controller does not observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible densities differ between members. SALS-adaptive additionally observes the realised biomass of the crop under the modules relative to an open reference plot, a quantity that can be measured on the farm by remote sensing or by a reference plot. A feedback rule that shares light whenever this ratio falls below a target serves as the hand-written counterpart; its target (1.0) was chosen on training-block seasons under the ensemble.

Uncertainty-aware design. The design rule of Section 2.4 chooses the largest GCR whose historical retention meets the floor. Under shade-response uncertainty we choose the largest GCR that meets the floor in at least 75% of five ensemble members, and compare it with a design under the ensemble mean and with an oracle design under the true response. Evaluation of this rule is leave-one-out: the truth is the held-out member and the design uses the other four.

Stress tests. Controllers and rules are also evaluated under four fixed responses, the proportional response, the response calibrated to the meta-analysis, the response fitted to the field trials and a harsh response (proportional growth and a 1.5 times larger harvest-index penalty). The proportional and harsh responses lie outside the range of the ensemble.

2.7. Evaluation protocol, statistics and upscaling

Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Training uses the 2001–2014 seasons of the training sites. Design selection uses the 2001–2014 seasons of the test sites. Evaluation uses the

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °C d, RUE in gMJ⁻¹; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- the mean electricity (MWh ha⁻¹ yr⁻¹) and the relative electricity e ;
- the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been

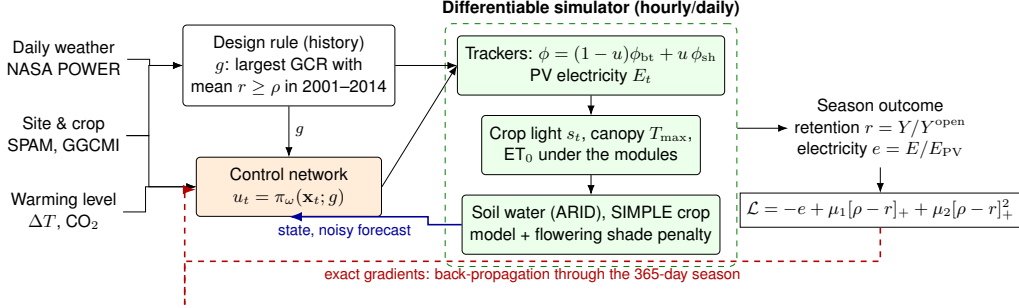


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

433 chosen from its 2001–2014 seasons with the same rule for all controllers. By
 434 construction all methods keep the mean yield retention close to 90% (0.904–
 435 0.917), so differences in electricity are differences in the density that each
 436 controller can sustain under this floor.

437 **Static arrays are the weakest.** Without light sharing, the floor limits
 438 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
 439 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 440 modules to let light through during the cropping season (seasonal shar-
 441 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule
 442 that shares light only while the crop is growing (phenology rule) reaches
 443 $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the
 444 crop on forecast heat or drought days (stress rule) is worse than the phenol-
 445 ogy rule (LER 1.18): its density is limited to 13% because it reduces light at
 446 times when the floor is most binding.

447 **The learned controller is best by a modest, consistent margin.**
 448 SALS sustains a density of 17% (similar to the phenology rule), produces
 449 $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Com-
 450 pared with the phenology rule this is +0.038 in LER (95% CI over sites
 451 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Ta-
 452 ble A.13). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35
 453 times the electricity at 2.9 times the density. The frontier in Fig. 6 shows
 454 the same picture at matched densities: the learned policy lies on or slightly

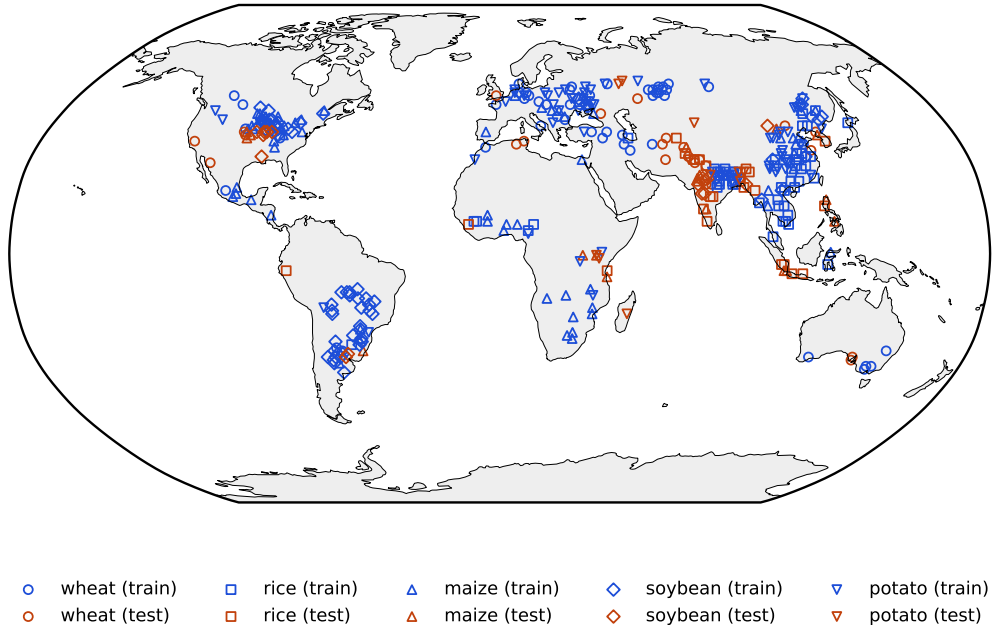


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at

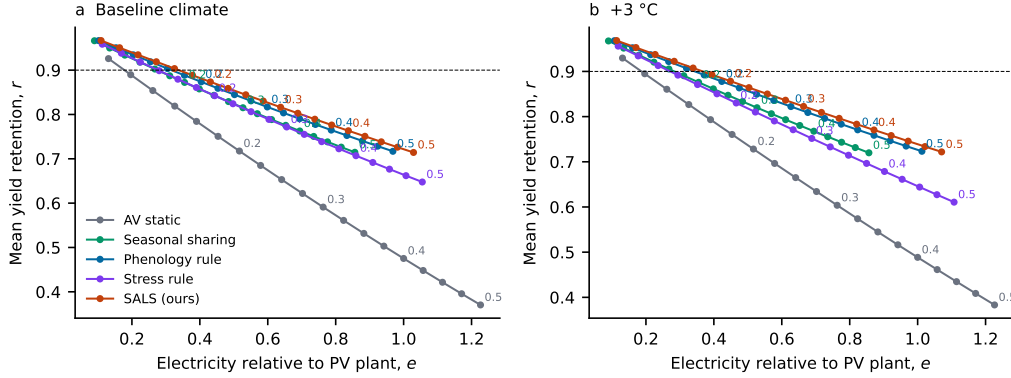


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

+3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and 0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at +3 °C).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET₀ forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

490 3.3. Floor conditioning, chance-constrained training and forecast errors

491 **One network for all floors.** Regulations and farmers set different
 492 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
 493 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
 494 design step for each floor (Table 3). Its LER exceeds that of the phenology
 495 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
 496 0.032 and 0.033 at $+3^\circ\text{C}$), so the benefit carries over to other floors without
 497 retraining. A network that received the floor as an additional input and was
 498 trained over all four floors did not improve on it (differences in LER ≤ 0.001).
 499 Because the design step already adapts the density to the floor, conditioning
 500 the controller on the floor is not needed.

501 **Chance-constrained training.** We also trained a network whose ob-
 502 jective penalises a batch share of seasons below the floor that exceeds 20%.
 503 It did not improve the tail. With the mean-based design step its LER was
 504 1.263 (baseline) and 1.290 ($+3^\circ\text{C}$) against 1.270 and 1.297, and with chance-
 505 constrained design it reached season-level compliance of 92.6% against 92.0%
 506 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
 507 by the design step (Section 3.1), not by the training objective.

508 **Forecast errors.** We raised the forecast errors seen by the controllers
 509 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
 510 of the phenology rule was identical at all four error levels to three decimals
 511 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
 512 1.416). Neither controller uses the weather forecast: both rely on the crop’s
 513 thermal time, which is observable from on-farm temperature records. The
 514 stress rule, which depends on the forecast of maximum temperature, degrades
 515 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 516 its LER from 1.432 to 1.418 as the errors triple.

517 **A tuned rule.** The phenology rule above was written, not tuned,
 518 whereas SALS is trained. We therefore tuned the window of the rule on
 519 the training-block seasons with the same objective (electricity at the density
 520 where mean retention equals the floor); the best window starts at 0.15 in-
 521 stead of 0.05 of the thermal-time requirement (Table 4). Pooled over three
 522 splits and 76 distinct crop–site pairs, the tuned rule raises LER over the orig-
 523 inal rule by 0.026 (95% cluster-bootstrap interval 0.016 to 0.037), and SALS
 524 exceeds the tuned rule by 0.018 (0.012 to 0.024), or 5.7% more electricity at
 525 equal retention, and the original rule by 0.044 (0.030 to 0.060). Tuning the
 526 rule thus accounts for about 60% of the margin of SALS over the written

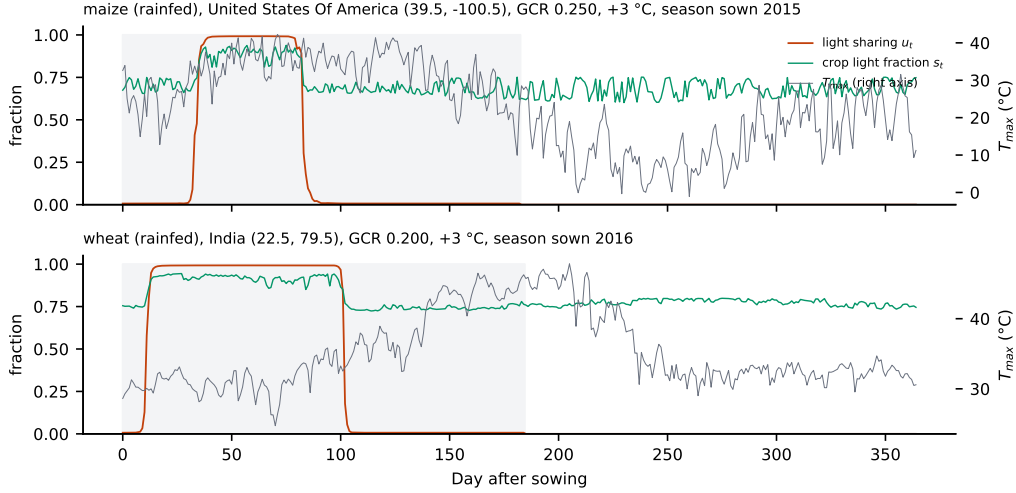


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

rule, and the remaining margin is small but not zero. SALS meets the floor in fewer seasons than the rules (83% against 84–85%).

3.4. What the learned policy does

Averaged over the rainfed test seasons of the main split at a density of 0.3, the light-sharing level of SALS is a smooth function of relative thermal time. It is close to zero in the first 10% of the thermal time required for maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau covers the window in which our crop model penalises shading through the harvest index. The ramps at both ends let the modules produce electricity when additional light is worth least to the crop. Fig. 4 shows two example seasons. The phenology rule, which is also the initialisation of the network, shares light during the whole growing period. Fine-tuning through the simulator found that sharing can be reduced at the edges of the season: SALS gains its advantage by yielding light when it matters and collecting electricity when it matters less.

The policy also shares less light on days with a forecast heat stress: the mean level is 0.35 on such days against 0.60 (baseline) and 0.51 ($+3^{\circ}\text{C}$)

on other days, and 58–63% of sites show a lower level on heat days. This cannot be read as a stress response, however. Heat days occur late in the season, when the phenological ramp is already falling, and a policy without any stress input shows a similar contrast (0.37 vs 0.57 in a separate check). The ablation above confirms that stress information does not improve the result.

3.5. *Crops and regions*

The benefit of light sharing differs between crops (baseline climate, SALS). Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20 for both). The stress rule has a lower LER for every crop (1.26 for soybean, 1.14–1.21 for the others).

The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose more than low-yielding ones. Regionally, Latin America and Oceania (4 sites) combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹) with a production-weighted retention of only 0.862. Africa and the Middle East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would therefore need a region- and crop-specific design step, not a global one.

3.6. *Model behaviour and validation*

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which

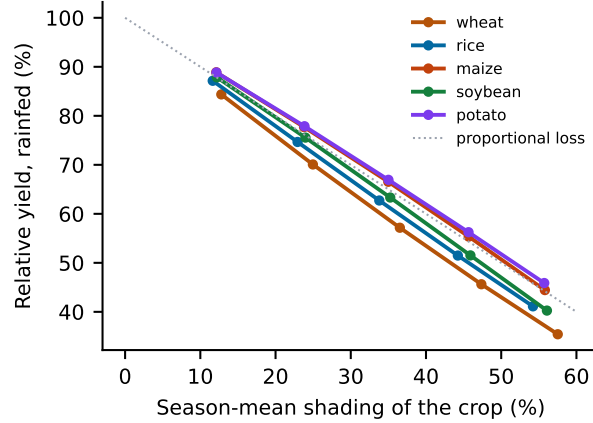


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 5) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 6 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19

602 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the
 603 order of the seasonal and stress rules changes. The absolute gains reported in
 604 this paper are thus lower bounds with respect to these trials and not upper
 605 bounds.

606 3.7. *A shade-response model learned from field data*

607 Cross-validation by study shows what the data can and cannot support
 608 (Tables 7 and 8). Over all 339 shaded observations the root-mean-square
 609 error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37
 610 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes
 611 from crop types that the simulator does not represent (berries, forages, fruits
 612 and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses,
 613 the learned curves are not better than proportional loss: the error is 0.222
 614 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$),
 615 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes
 616 and tubers. Partial pooling is as good as separate curves (0.312 against 0.313)
 617 and the covariates add almost nothing (0.311). Gradient-boosted trees on the
 618 same inputs are worse (0.321) than the parametric model. The error of 0.22
 619 for the four crop types is large: yields of a new study differ from the learned
 620 curve by about 20 percentage points, which is the between-study spread
 621 that a design under uncertainty has to cover. Crop type is the information
 622 that transfers between studies, and none of the covariate effects on $\log \beta$ is
 623 clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ;
 624 subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45).
 625 The estimated effect of solar-panel shade is not interpretable, because all 46
 626 panel observations are of leafy vegetables, mainly lettuce, and the effect is
 627 confounded with crop type.

628 The learned curves show a clear ordering of crop types (Table 9). At
 629 20% shading C3 cereals and tubers lose less than proportionally (relative
 630 yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40%
 631 shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide
 632 for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The
 633 simulator’s responses are close to the data for cereals and potato but, under
 634 both calibrated ensembles, too mild for maize and soybean at 40% shading
 635 (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration
 636 cannot lower the maize response further, because the simulated shading also
 637 relieves heat and drought stress and the compensation and harvest-index pa-
 638 rameters are bounded below at zero. The field trials of maize (Ramos-Fuentes

et al., 2023) are also milder than the meta-analysis. For maize and soybean the gains reported in this paper are therefore probably overestimated.

3.8. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the five crops to the data-driven curves and the field trials (ensemble E2, Table 10). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the largest compensation (0.91 ± 0.31) and the largest uncertainty.

Learned controllers. Controllers were trained on E1 and evaluated under four fixed responses on 21 crop-site pairs of split 0 with one training seed, so the intervals below, which are cluster bootstraps over pairs, do not include seed or split variation. Under every response SALS stays better than the written phenology rule (Table 11): by +0.019 (0.007 to 0.039), +0.024 (0.009 to 0.049), +0.038 (0.017 to 0.062) and +0.015 (0.001 to 0.038) LER under the proportional, calibrated, field-fitted and harsh responses. Training on the ensemble did not help: SALS-robust differs from SALS, which was trained on the proportional response only, by -0.005 to $+0.003$ LER with every interval containing zero, and SALS-adaptive, which sees the measured biomass ratio, by -0.010 to $+0.007$ with every interval containing zero. The training ensemble E1 is narrow, so this null result may reflect its low diversity. The feedback rule is worse than the phenology rule by 0.026–0.058 LER under every response (intervals exclude zero). A plausible reason is that the biomass ratio does not reveal the harvest-index loss from shading around flowering, but we did not test this explanation.

Design under uncertainty. The larger risk is in the design step (Table 12, Fig. 6). Pooled over the three spatial splits (76 distinct crop-site pairs), a density chosen under the mean response of E2 leaves 69% of the pairs at the floor when the true response is another plausible one, against 87% for an oracle design; with the wider ensemble E3 the figures are 70% and 85%. Choosing the density so that the floor is met in at least 75% of the plausible responses raises compliance by +9.0 points under E2 (95% cluster-bootstrap interval 5.3 to 12.9) and +8.7 under E3 (4.2 to 13.7), at a cost of 0.038 and 0.034 in LER; it recovers about half of the gap to the oracle. The ensemble-based rule is, however, not better than a simple margin. Keeping the density rule of the main analysis and raising the floor used in the design

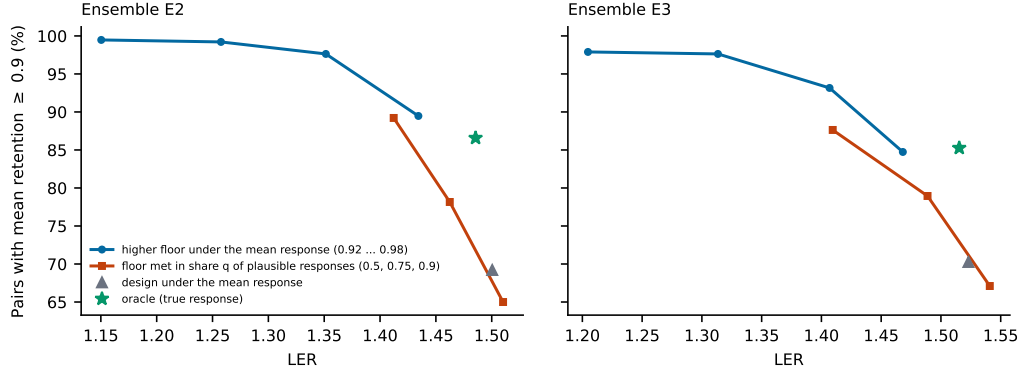


Figure 6: Share of crop-site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue), a floor met in a share q of the plausible responses (orange), the design under the mean response and the oracle design.

from 0.90 to 0.92 under the mean response gives 89% (E2) and 85% (E3) of the pairs at the floor at LER 1.434 and 1.468, whereas the ensemble rule at 90% reliability reaches 89% and 88% at 1.412 and 1.409: at equal compliance the margin keeps 0.02 (E2) to 0.04 (E3) more LER. The ensemble is thus useful to size the margin (about 0.02 to 0.04 above the floor for 85–95% of the pairs) but a rule built on its quantiles does not beat it. On the first split alone, with the earlier ensemble E1, the quantile rule restored the oracle level (76% and 74%, against 58% and 55% under the mean response), but without the margin comparison. The guarantee holds only within the range of responses that the ensemble spans: with the E2 design on split 0 (21 pairs), compliance under the proportional, calibrated and harsh responses is 19–24% (mean-response design 19%), and 81% under the field-fitted response (71%).

3.9. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 7) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize −31%, rice −27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265),

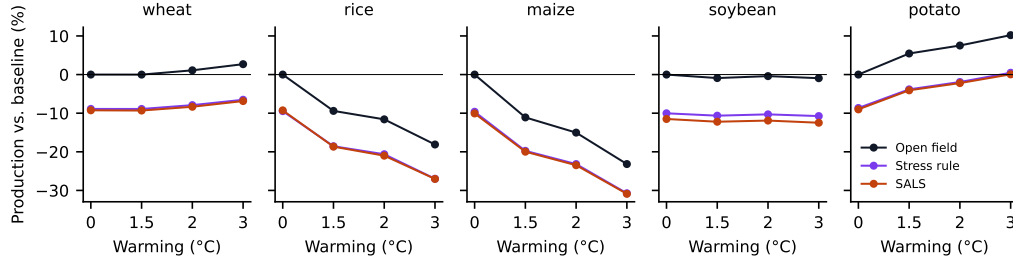


Figure 7: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table A.13).

3.10. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr⁻¹ under the baseline climate and 16.0 PWh yr⁻¹ at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr⁻¹, and static arrays 6.4 and 6.9 PWh yr⁻¹. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 8): the best 2% of the staple cropland would supply 16 PWh yr⁻¹ and the best 5% 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

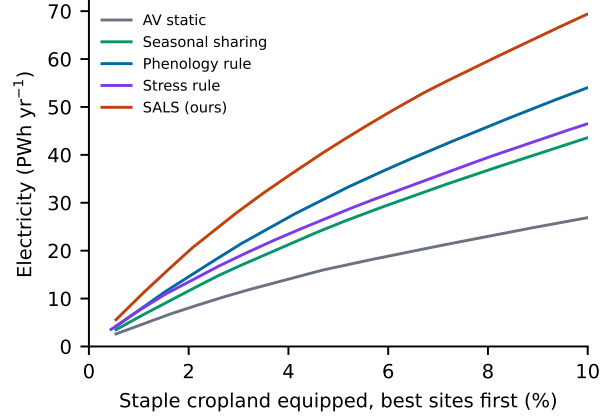


Figure 8: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

723 4. Discussion

724 4.1. What the results establish

725 Four findings stand on consistent evidence. First, sharing light with the
726 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
727 can produce under a yield floor. With static trackers the floor limits arrays
728 to about 6% ground coverage and an LER of 1.07. Any light-sharing
729 control allows about three times the density, and the LER rises to 1.18–1.24
730 for hand-written rules. Second, the best simple rule follows crop phenology.
731 It shares light across the middle of the season, around flowering and grain
732 filling, and produces electricity at the beginning and the end. It needs only
733 the crop’s thermal time, which is observable on the farm, and it is insen-
734 sitive to forecast errors, whereas the rule that reacts to heat and drought
735 stress is worse in every climate and degrades with forecast error. Third,
736 the qualitative result does not depend on the weakest part of the model.
737 The simulated shade response is conservative against the field trials we could
738 use (Section 3.6); calibrating it raises the allowed density, and the phenol-
739 ogy rule stays the best method, with a gain over static arrays of 0.17, 0.19
740 and 0.30 LER under the conservative, calibrated and field-fitted responses.
741 However, the response learned from the meta-analysis shows steeper losses
742 for maize and grain legumes than the simulator reproduces (Section 3.7),

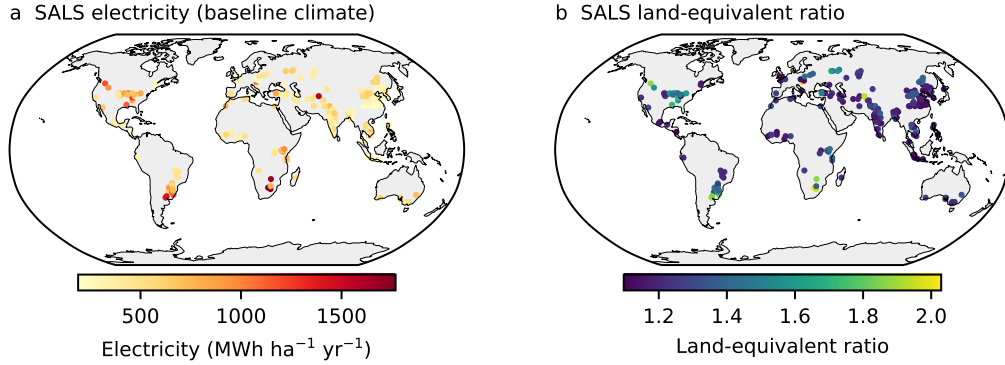


Figure 9: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

so the absolute gains are lower bounds for cereals and potato and probably overestimates for maize and soybean. Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.04 in LER against the written rule (0.018 against a rule whose window was tuned on the training seasons, 5.7% more electricity at equal retention), reproducibly over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and at 84–86% of the held-out sites. This gain is small compared with the gap between static and dynamic control.

4.2. What role does artificial intelligence play?

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. The learned model is useful mainly as a measurement of how little can be learned: a hierarchical model fitted to 428 observations from 58 studies predicts a held-out study of the four staple crop types with an error of 0.22 in relative yield, no better than proportional loss, and the gain over proportional loss (0.43 to 0.31 over all types) comes from crop types that the simulator does not represent. Crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). Two things follow. The shade response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve, and the process-based simulator does not reproduce the steep losses of maize

766 and legumes seen in the data. Learning the response was limited by the data
767 and not by the model: two studies support the tuber curve, no rice study
768 exists, and 46 panel observations are all leafy vegetables.

769 The second is calibration through the differentiable simulator. Gradient
770 descent through whole seasons fitted the response of each crop to the learned
771 curves and to the field trials, and it produced an ensemble of plausible re-
772 sponses that reconciles them (Section 3.8). The ensemble informs the design
773 decision: a density chosen under the mean response leaves about 70% of
774 crop–site pairs at the floor if another plausible response holds, against 85–
775 87% for an oracle, and a density chosen to meet the floor in 75% of plausible
776 responses recovers about half of that gap at a cost of 0.04 LER. A simple
777 margin, however, does better: raising the floor from 0.90 to 0.92 under the
778 mean response reaches 85–89% of the pairs and keeps 0.02–0.04 more LER
779 than the ensemble rule at equal compliance. What the ensemble adds is the
780 size of the margin, not a better rule. The bottleneck of tracker-based agri-
781 voltaics under a food-security floor is the uncertainty of the crop response
782 that enters the design, and it is data on that response, not larger networks,
783 that would improve the result; the learned curves show a spread of about
784 20 percentage points between studies. The guarantee holds only within the
785 range of responses that the ensemble spans.

786 The third is control, and here learning contributes little. The learned
787 controller adds 0.018 LER over a tuned phenology rule (and 0.044 over the
788 written one), and 0.01–0.03 over the written rule under the alternative re-
789 sponses on one split; a reader who needs a simple, auditable controller can use
790 that rule and obtain most of the electricity gain over static arrays. Neither of
791 the two extensions that we expected to matter under uncertainty did: train-
792 ing on an ensemble of responses did not change the policy, and neither did
793 giving the controller a farm-measurable signal, the biomass of the crop rela-
794 tive to a reference plot, nor a feedback rule built on that signal. The learned
795 policy also has to be initialised from the phenology rule (Section 4.5). We
796 therefore do not claim that learned control is what makes agrivoltaics work:
797 in this model, the schedule that a learned policy converges to is a smooth
798 function of crop phenology, which a rule can already express.

799 4.3. *What the results do not establish*

800 The early hypothesis that stress-aware control, that is, shading the crop
801 on heat and drought days, would be a main source of benefit under warming
802 was not supported. The stress rule performs worse than the phenology rule

803 in every climate, and a learned controller without stress inputs performs as
 804 well as one with them. A plausible reason is that in our crop model the yield
 805 gain from shading on a hot or dry day is smaller than the loss of light on
 806 that day, so stress information does not help. Whether this holds for crop
 807 models with more detailed canopy-temperature and heat-stress responses is
 808 an open question. Agrivoltaics does not protect staple production against
 809 warming in our simulations: light-sharing systems produce about 90% of the
 810 open-field yield in every climate and follow the same decline in maize and
 811 rice (Fig. 7).

812 4.4. Food security in practice

813 Holding the historical mean retention at 90% does not hold it in each
 814 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
 815 slightly heavier than for the rules, and production-weighted retention is below
 816 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
 817 constrained design raises season-level compliance to about 92% at a cost of
 818 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
 819 mulation is chosen, a per-site regulatory floor would need a region- and crop-
 820 specific design step. The upscaled potential (Section 3.10) shows the trade-off
 821 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
 822 need about 10 TWp of modules, several times the present global fleet, and
 823 the best sites are rainfed maize and soybean areas where production-weighted
 824 retention is lowest.

825 4.5. Limitations

- 826 • *Simulation only.* All results come from a simulator. PV output was
 827 benchmarked against PVGIS (29 sites, mean bias −4.9%), but the crop
 828 response to shade was learned from one meta-analysis data set and
 829 compared with four field trials (Sections 3.6 and 3.7). The simulated
 830 response is conservative with respect to the trials (rice, wheat, potato,
 831 maize) but too mild for maize and soybean relative to the meta-analysis,
 832 where the 40%-shading relative yields are 0.42 and 0.40 against 0.75
 833 and 0.48 in the simulator. The harvest-index sensitivities h are assump-
 834 tions, and the sensitivity analysis uses rule controllers on a subsample
 835 of 22 sites.
- 836 • *Canopy cooling.* The effect of shade on canopy temperature was repre-
 837 sented by one coefficient κ , applied to a heat-stress function that was

calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.

- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency. More detailed models with grain-number formation and canopy energy balance may favour different schedules.
- *Climate scenarios.* Warming was applied as uniform temperature shifts with fixed precipitation, radiation and relative humidity, and with unchanged cultivars and sowing dates. It is not a bias-adjusted climate projection.
- *Initialisation and state information.* The network is initialised from the phenology rule, because training from random weights collapsed to full sharing. The gain of SALS is therefore the gain of fine-tuning that rule, and we cannot say whether a better optimiser or a richer policy class would find larger gains. The controller uses the simulated state of the crop and soil (thermal time, canopy interception, soil water) without measurement error, and the forecast errors we tested apply to weather only.
- *Economics and deployment.* There is no cost, market or grid analysis. The value of midday electricity, curtailment, storage, structural cost and farmer acceptance (Trommsdorff et al., 2021) will determine the realisable share. The average row pitch at the chosen densities (about 13 m for 2.2-m modules) leaves room for machinery, but the densest designs (GCR up to 0.5) have a pitch of only 4.4 m.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever of tracker-based agrivoltaics under a food-security floor. In a differentiable simulator of trackers, PV output, crop light, microclimate and crop growth, evaluated fairly on 400 crop-site pairs with the array density of each site chosen from its own history, light-sharing rules allowed about three times the module density of static arrays at a mean yield retention of 90%. They raised the land-equivalent

ratio from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no weather forecast was the best of them. The simulated shade response is conservative against the field trials we could use, so these gains are lower bounds, and calibrating the response raised them without changing the ranking of methods. The crop’s response to shade, the least certain input, was learned from 428 field observations: a hierarchical model predicted a held-out study of the staple crop types no better than proportional loss (error 0.22), which shows the size of the between-study spread, and gradient descent through the simulator turned it into an ensemble of plausible responses. The simulator is too mild for maize and soybean, so the gains for those crops are probably overestimated. A learned controller exceeded the written phenology rule by 0.044 LER and a rule with a window tuned on the training seasons by 0.018 (95% interval 0.012–0.024), by following a smooth phenology-dependent schedule that a rule can express; training it on an ensemble of shade responses or giving it a farm-measurable biomass signal added nothing. The decision that mattered was the design. A density chosen under the mean response left about 70% of crop–site pairs at the floor when another plausible response held (oracle 85–87%); choosing the density so that the floor holds in 75% of the plausible responses recovered about half of the gap at a cost of 0.04 LER, but a floor raised by 0.02 under the mean response did better at equal compliance. The ensemble is therefore useful to size a safety margin, not to replace it. The yield floor is met on average but not in every season or region, and the electricity potential of 5% of staple cropland (about 14 PWhyr^{−1}) requires about 10 TWp of modules. Future work should validate the crop response to shade under trackers with more field data, notably for maize and legumes where the meta-analysis and the trials disagree, extend the crop model to grain-number formation and canopy energy balance, add costs and grid constraints, and pursue controllers that guarantee the food-security floor per site and per season.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

904 **Declaration of competing interest**

905 The author has no known competing financial interests or personal re-
906 lationships that could have appeared to influence the work reported in this
907 paper.

908 **Data and code availability**

909 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
910 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
911 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
912 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
913 ments, together with the site list, is available from the author’s public reposi-
914 tory (<https://github.com/vinhqddang/Artificial-Intelligence-in-Agriculture>).

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918 **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Rules and the learned controller pooled over the three spatial splits (76 distinct crop-site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05). The tuned phenology rule shares light while thermal time lies between 0.15 and 1.0 of the season requirement (window chosen on training-block seasons; the original rule uses 0.05 to 1.0).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule	0.154	0.292	0.916	85	91	1.207
Tuned phenology rule	0.161	0.317	0.916	84	89	1.233
SALS	0.161	0.335	0.916	83	87	1.251

Table 5: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 6: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 7: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 8: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 9: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 10: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 11: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out site, 21 sites). The proportional (conservative) and harsh responses lie outside the range of the calibrated ensemble.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust (ensemble-trained)	1.179	1.204	1.491	1.127
SALS-adaptive (ensemble, biomass feedback)	1.176	1.202	1.495	1.124

Table 12: Array design under shade-response uncertainty, phenology rule, pooled over three spatial splits (76 distinct crop–site pairs; every third held-out pair). Truth: each of five members of the ensemble in turn; designs of the first group use the other four members (leave-one-out). Ensemble E2 is calibrated to the learned curves and field trials, E3 additionally carries the between-study spread (target errors 0.2). Pairs ≥ 0.9 : share whose mean retention reaches the floor. The last three rows keep the density rule of the main analysis but raise the floor used in the design.

Design	E2		E3	
	Pairs ≥ 0.9 (%)	LER	Pairs ≥ 0.9 (%)	LER
oracle (true response)	87	1.486	85	1.515
mean response	69	1.500	70	1.523
floor met in 50% of responses	65	1.510	67	1.541
floor met in 75% of responses	78	1.462	79	1.489
floor met in 90% of responses	89	1.412	88	1.409
floor 0.92 under mean response	89	1.434	85	1.468
floor 0.94 under mean response	98	1.351	93	1.407
floor 0.96 under mean response	99	1.258	98	1.313

Table A.13: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92